

500 Sets

Elementary – Science

1. Currents named for this type of heat transfer exist in the Earth's mantle. A certain type of oven is known for using this form of heat transfer. Give this form of heat transfer that involves travel of molecules through fluids.

CONVECTION

2. Much of the earth's electric power comes from steam turning the wheels of this type of machine. A 'greener' alternative power source comes from these three-bladed modern windmills, designed for really breezy areas. Name the type of engine that, appropriately, gets its name from the Latin for "spinning."

TURBINE(S)

3. At this point, evaporation occurs. A physical change occurs, such as when steam comes off from a pot of hot water. Identify this point which, for water, is 100 degrees Celsius?

BOILING (POINT)

4. The speed of this force is measured by an anemometer [ann-ih-MOMM-ih-ter]. A weather vane tells its direction. Name this weather phenomenon, a major component of hurricanes and tornadoes.

WIND

5. They weigh approximately the same as protons, which are also located in the nucleus. Added to the protons, they give atomic weight. What are these subatomic particles with no charge?

NEUTRON(S)

6. One type of these devices are found in intensive-care units, and provide special care to premature babies. All of its types are used to provide care for young or unborn animals. What are these devices, one version of which warms eggs before they hatch?

INCUBATOR(S)

7. The blood transports this gas from the lungs throughout the body. All vertebrates require this gas in order to breathe. What gas has an atomic symbol of capital O?

OXYGEN

8. This man's most famous quote is thought to be missing the word "a." Along with Buzz Aldrin and Michael Collins, he led the Apollo 11 mission. Name this astronaut, the first human to set foot on the moon.

(NEIL) ARMSTRONG

9. This quantity might explain why the universe is expanding faster and faster. Along with dark matter, it is thought to make up 95 percent of the matter and energy in the universe. What is this proposed quantity, thought to be pushing galaxies apart from one another?

DARK ENERGY

10. She is the only woman who has won two Nobel prizes. Along with her husband, she discovered several radioactive elements. Who is this Polish physicist, who did research with her husband Pierre?

(MARIE) CURIE

11. These marine animals come in colossal and giant varieties, which dwell in the deep ocean. Along with octopuses, they are the best known cephalopods. What are these animals, sometimes eaten by humans as calamari?

SQUID(S)

12. They are one of the only known mammals to have venom. Along with the echidnas [ee-KID-nuhs], they make up the monotreme order. What are these semi-aquatic, duck-billed animals?

(DUCK-BILLED) PLATYPUS

13. You get one of these when you combine two substances, and each substance retains its own chemical identity. An example of this includes combining flour and sugar together. What is this result of combining substances, that can be classified as homogeneous or heterogeneous?

MIXTURE

14. This helps regulate air temperature by absorbing thermal radiation. As it increases, the potential for storms also increases. What do we call the amount of water vapor in the air?

HUMIDITY

15. This kind of cloud is composed of ice crystals which display as white patches or narrow bands. At sunrise and sunset, these clouds often display a bright yellow or red color. What are these clouds with delicate filaments called?

CIRRUS [SIR-uhs] (CLOUDS)

16. For ice, this temperature level is 32 degrees Fahrenheit. At what point does a substance turn from solid to liquid?

MELTING (POINT)

17. This property can be measured by the Mohs Scale. Based on that scale, it is highest for diamond and lowest for talc. What is this property that measures how much a substance gives way to pressure?

HARDNESS

18. Galvanization can prevent this type of disintegration from occurring, preventing the formation of iron oxide. By what process does iron acquire a reddish tint after being exposed to moist air or water for a long time?

RUSTING (Accept: OXIDATION and other forms of either word)

19. In many amusement park rides, those on the ride will encounter centripetal force. Centripetal force will always be pointed towards the center of motion. What is the type of path an object will follow if it is experiencing a centripetal force?

CURVED (ACCEPT: CIRCULAR or SIMILAR RESPONSES))

20. If you study smelt, pilchards, and bream, you are one of these scientists. Children who have an aquarium at home just might grow up to become one. Identify the 13-letter word for a person who studies fish.

ICHTHYOLOGIST

21. This organ connects directly to the brain stem. Damage to it can result in paralysis. Identify this organ, which along with the brain, makes up the central nervous system.

SPINAL CORD (ACCEPT: SPINE)

22. Scientific name, please. The vaccination known as a DPT protects against diphtheria, pertussis, and this ailment, which causes sustained muscle contraction. Deep cuts and puncture wounds, especially from metal, are likely to lead to what disease, also known as lockjaw?

TETANUS

23. This kind of landform ejects a porous rock called pumice. Depending on how quickly cooling occurs, it can become a shield or cone shape. What kind of eruptive landform ejects lava?

VOLCANO

24. A molecule of water has three of these, two hydrogen and one oxygen. Each of these consists of protons, electrons, and sometimes neutrons. What is considered the smallest unit of a chemical element?

ATOM

25. Examples of these kinds of molecules include glycine [GLIGH-seen] and methionine [meh-THIGH-uh-neen]. Each one contains a central carbon atom, and has a particular side chain. What molecules are the building blocks of proteins?

AMINO ACIDS

26. This layer of the atmosphere is able to reflect radio waves. Energy from the sun leads to its namesake particles, as well as free electrons in this layer. What is this layer of the atmosphere that contains many charged particles?

IONOSPHERE

27. These rocks can be divided into intrusive and extrusive varieties. Examples include basalt and obsidian. What are these rocks formed through volcanic activity?

IGNEOUS [IG-nee-us] (ROCKS) (Not LAVA)

28. This informal group of animals have no notochord. Examples include comb jellies, sponges, and arthropods. Identify this type of animal, named for their lack of a backbone.

INVERTEBRATE(S)

29. An animal described by this term might eat both papaya and bush babies. Examples of these animals include catfish, ants, dogs, and humans. What is this term for animals that consume both plants and animals?

OMNIVORES (Accept: OMNIVOROUS)

30. Coding systems can be used to convert non-numerical data to this type of data. Examples of this type of data might include mass, height, time, or number of events. Name this type of data, often contrasted with qualitative data.

QUANTITATIVE (DATA)

31. This general form of reproduction is most often seen in simple lifeforms such as bacteria and sponges. Examples of this type of reproduction include budding, binary fission, and fragmentation. Identify the form of reproduction that creates a genetic copy of an organism.

ASEXUAL

32. Genes that follow a pattern of dominant or recessive inheritance are named after him. Experiments with pea plants made famous what Austrian monk?

(GREGOR) MENDEL

33. Parts of them include the chorion, amnion, and yolk. Fish and amphibians lay them in water, while reptiles and birds lay them on land. What is this name for the female reproductive product?

EGG

34. This astronaut was the youngest American to have ever traveled to space. Following in the steps of two cosmonauts, she was the third woman in space. Who was this first American woman to travel to space?

(SALLY) RIDE

35. Because of air resistance, a freely-falling object will increase in velocity until it reaches this speed. For a human, this value is about 120 miles per hour. What is this value, the highest velocity that an object can have when falling in air?

TERMINAL VELOCITY (ACCEPT: TERMINAL SPEED)

36. This is an example of a simple machine. For this simple machine, the fulcrum, or pivot point, often occurs between the load and the effort. What is this simple machine, of which a see-saw is an example?

LEVER

37. It is sometimes known as the law of inertia. Friction is one force that needs to be factored in when considering this law. What is this law of motion that states that an object in motion will stay in motion and an object at rest will stay at rest, unless acted on by an outside force?

(NEWTON'S) FIRST (LAW OF MOTION) (Accept: ONE)

38. This term used in astronomy, beginning with "B", actually refers to two objects traveling together. From the Latin for "two together," it is a pair of stars that revolve around one another. Give the term for these double star systems.

BINARY (STAR) (SYSTEM)

39. These areas tend to form because the sun's radiation does not hit them as directly as at the equator. Give the areas found on Mars and Earth where large amounts of solidified water exist in sheets.

(POLAR) ICE CAPS (Accept: GLACIERS)

40. This organ is found in birds and reptiles, and in many mammals is controlled by the diaphragm. Give the name of the organ that allows the expulsion of carbon dioxide and the intake of oxygen.

LUNG

41. This part of plant anatomy contains the ovary, style, and the stigma, which receives pollen. Give the name of this part of the flower, the female reproductive organ.

PISTIL (ACCEPT: CARPEL)

42. This rock is an example of chert, which is a type of quartz. Give the name of this rock, which is often associated with weapons and starting fires.

FLINT

43. The males of this insect are known for their very loud songs by vibrating membranes on their bodies. Give the name of this type of insect that can appear yearly, every 13 years, or every 17 years.

CICADAS (Not: LOCUSTS)

44. The one representing July is ruby, and the one representing February is amethyst [AM-uh-thist]. Give the term for a mineral traditionally assigned to each month of the year.

BIRTHSTONE

45. In astrology, twelve of these astronomical arrangements determine an individual's fate. Give the term for arrangements of stars into shapes, like the twins Gemini or Cancer the crab.

CONSTELLATIONS

46. On a forecast map, these are represented by red, blue and sometimes purple lines, with triangle and semicircle patterns. Give the term for boundaries between air masses that can be cold or warm.

(WEATHER) FRONTS

47. This feature of our atmosphere is usually between six and nine miles high, and often forms in the boundary between cold and warm fronts. Give the term for strong currents in earth's troposphere through which airplanes travel.

JET STREAM(S)

48. Bacteria and fungi serve as these kinds of organisms in the food chain, adding nutrients to the soil in the process. Give the term for the kind of organism that breaks down dead or decaying plants and animals.

DECOMPOSER (Accept: DETRITIVORE, SAPROBE, SAPROPHYTE, or SAPROPHAGE)

49. This component of living things may contain smaller functional parts, like ribosomes and chloroplasts, called organelles. Give the term for the smallest biological unit, or basic structural unit, for all living organisms.

CELL

50. This process begins with prophase and ends with telophase. Give the term for the splitting of a cell.

MITOSIS

51. Many types of this exist, including marine, freshwater, and desert. Give this term for all of the organisms, and their surroundings, in a particular area of earth.

ECOSYSTEM

52. He became an oceanographer after an accident meant he could not become a navy pilot. He co-invented the Aqua-Lung. What scientist had a television series called the "Undersea World"?

(JACQUES) COUSTEAU

53. This chemist, engineer, and industrialist worked under John Ericsson in the U.S. before the Civil War. He invented dynamite and other powerful explosives. Name the Swede who founded the world's most prestigious awards given for intellectual achievement, which bear his name.

(ALFRED BERNHARD) NOBEL

54. The four largest moons of Jupiter were discovered by this man. He is also known for forwarding Copernicus's heliocentric model. Name this Italian astronomer, known for his telescope.

GALILEO GALILEI (Accept either or both names)

55. This scientist was the first to use a prism to divide light into its parts. He is credited with discovering calculus, and he did research on optics. Identify this scientist, known for his laws of gravity and of motion.

(SIR ISAAC) NEWTON

56. This scientist was an early pioneer of alternating current. He is known for a namesake coil and was a rival of Thomas Edison. Who is this scientist, the namesake of a car company owned by Elon Musk, which produces the Model S?

(NICOLA) TESLA

57. This American created a type of stove, as well as bifocal lenses. He is perhaps best known for his lightning rod, and is often credited with discovering electricity. Who is he?

(BENJAMIN) FRANKLIN

58. His research helped show that life could not come from non-living materials. He is sometimes called "the father of microbiology," and he contributed to the development of vaccines. Who developed a process which is now used to sterilize milk and other dairy products?

(LOUIS) PASTEUR [pas-TUR]

59. George Washington Carver devised many uses for several plants, including this one. He testified to Congress on the importance of creating a tariff on importing these nuts. What nut often sold at baseball games did Carver study extensively?

PEANUT

60. This scientist is known as the "father of immunology". He was a pioneer of a vaccine for a disease that has killed millions of people. Who is this scientist, that is famous for his discovery of the smallpox vaccine?

(EDWARD) JENNER

61. A type of radiation that comes from black holes is named for him. He wrote A Brief History of Time and is known for having ALS. Who is this scientist that died in 2018, known for using a computerized voice system?

(STEPHEN) HAWKING

62. These particles are made of quarks and weigh around the same as neutrons. Helium has two of these and hydrogen has one. What are these positively charged subatomic particles?

PROTONS

63. While cell membranes might be around every cell, this is only found around plant cells. Helping plants keep their shape, name the protective coating around plant cells that is made of cellulose.

CELL WALL(S)

64. This scientist and author helped inspire the creation of the EPA. Her most famous work detailed the effects of DDT. Name this author of Silent Spring.

(RACHEL) CARSON

65. This Italian astronomer and mathematician's work and teaching helped experimentation to become a recognized method for discovering facts of nature. His discoveries with the telescope revolutionized astronomy. Name this man whose discoveries paved the way for the acceptance of Copernicus' heliocentric system which set the sun at the center of the solar system.

GALILEO (GALILEI)

66. This scientist studied finches and sailed on the Beagle. His most famous work was "On the Origin of Species." Who is this scientist, known for his theory of evolution by natural selection?

(CHARLES) DARWIN

67. According to the American Academy of Pediatrics, children should not spend more than this amount of time on electronic devices each day. How much time should be the maximum number of hours spent on computers, watching TV, and playing video games daily?

2 (HOURS)

68. Formal name, please. This star is actually a triple star in the constellation Ursa Minor. Identify earth's current North Star.

POLARIS

69. These dark clouds often produce light precipitation, like drizzle. Identify the blanket-like clouds that are the lowest of all clouds.

STRATUS (CLOUDS)

70. Examples of this kind of rock include marble and slate. Identify the kind of rock formed over time by pressure and heat.

METAMORPHIC

71. Cooler areas on this layer of the sun are called sunspots. Identify the layer of the sun that is its visible surface.

PHOTOSPHERE

72. During this phase of the moon, the moon is between the Earth and the Sun. Identify the phase of the moon during which the moon is not visible from Earth, marking the start of a new lunar cycle.

NEW MOON

73. These types of clouds are vertical and appear during fair weather. Identify the puffy clouds that look like cotton balls.

CUMULUS (CLOUDS) (Accept: CUMULI)

74. This temperature scale is based on the way water acts in its environment, and its two reference points are 100 degrees apart. Identify the temperature scale with water's freezing point at 0 degrees and its boiling point at 100 degrees.

CELSIUS (SCALE) (Accept: CENTIGRADE)

75. These ancient objects can be preserved in crystal, amber, or rock. Identify these items, which are the remains of previously living organisms.

FOSSIL(S)

76. Examples of these include the MMR and one for chickenpox. Identify these medical procedures that help prevent viral infections, usually given in shots.

VACCINE(S)

77. He proposed an idea of the subconscious that included id, ego, and super-ego components. Identify this Austrian psychologist, known for writing about the interpretation of dreams.

(SIGMUND) FREUD [FROYD]

78. Examples of them include scorpions, mites, and spiders. Identify this class of arthropods, which have eight legs.

ARACHNID (ACCEPT: ARACHNIDA)

79. She is often called the founder of modern nursing. Identify this English reformer, who is often known as "The Lady with the Lamp."

(FLORENCE) NIGHTINGALE

80. It is orbited by Charon [KAIR-un] and is the largest dwarf planet by volume. Identify this former planet, named for the Roman equivalent of Hades.

PLUTO

81. This is the largest component of the integumentary system, and modified types of this form the teeth and nails. Identify this largest organ of the human body.

(THE) SKIN

82. This numbered law can be written as F equals M times A . F is force, M is mass, and A is acceleration. Identify this law put forward by Isaac Newton, sometimes known as the law of acceleration.

(NEWTON'S) SECOND (LAW OF MOTION) (Accept: 2)

83. This astronomical event occurs after the waning crescent, and is only visible from Earth during a solar eclipse. Identify this phase of the moon, notable for the absence of a visible moon, which occurs at the start of each moon cycle.

NEW (MOON)

84. This process involves the transfer of a certain substance from the stamen to the stigma. Identify this process, that is often aided by bees and other insects, and is essential for plant reproduction.

POLLINATION

85. This quantity is represented by the letter "C" in a notable Einstein equation. Identify this quantity, roughly equal to 186,000 miles per second.

(THE) SPEED OF LIGHT

86. Had this scientist developed his work earlier, Franklin Roosevelt may not have been wheelchair bound. Identify this scientist who created the polio vaccine.
- (JONAS) SALK
87. This stage of development occurs after the embryo and the zygote. Identify this term, referring to an unborn human during pregnancy.
- FETUS(ES)
88. In 1938, a DuPont chemist's attempt to make a gas refrigerant didn't pan out, but it did lead to the creation of this fluorocarbon. Identify this trademarked, solid, non-stick polymer used to coat the surfaces of cookware.
- TEFLON
89. This climate is characterized by large amounts of rainfall for much of the year and warm temperatures year round. Identify this type of climate, typical of areas near the equator.
- TROPICAL (CLIMATE)
90. Examples of this type of rock include slate and marble. Identify this type of rock, created when other rocks are transformed under heat and pressure.
- METAMORPHIC (ROCK)
91. Some components of these are known as molecules or formula units, and examples include water and C-O-2. Identify this type of substance, which is made up of a combination of elements.
- (CHEMICAL) COMPOUND
92. During this part of the plant life cycle, the seed does not grow. If the seeds are kept cool and dry, it may last for years. During what period can the seeds be said to be sleeping?
- DORMANT (Accept other forms of the word)
93. This order of mammals, whose greatest number of species live in the South Pacific, includes the Tasmanian devil and wombat. Immediately after birth, the tiny, blind infants climb up their mother's belly, and remain for several months in her pouch. Identify this group of animals, which also includes the opossum, koala, and kangaroo.
- MARSUPIAL(S)
94. This names the protoplasm of a cell within the cell membrane, but excluding the nucleus. In cells having a nucleus, this contains all of the organelles [awr-guh-NELS] including the mitochondria [mahy-tuh-KON-dree-uh]. What do we call this semifluid substance?
- CYTOPLASM [SAHY-tuh-plaz-uhm]
95. This scientist lends his name to a very cold state of matter, along with Bose. In his best known equation, energy is set equal to mass times a factor of the speed of light. Name this scientist, best known for his equation $E = mc^2$.

(ALBERT) EINSTEIN

96. This term refers to an object's lack of transparency. In paper, this quality refers to the amount of "show-through" of an image from one side of the paper to the other. What is this term that refers to the state or quality of being opaque?

OPACITY [oh-PAS-i-tee]

97. In bacteria, chemicals in this structure separate different types of bacteria. In plants, they are made of cellulose. Identify this structure that provides extra support for a cell, around the membrane.

(CELL) WALL

98. This is the most abundant element in the entire universe. In stars, it is fused into helium, which leads to the release of energy. Name this element with atomic number 1 and symbol H, and which helps make up part of a water molecule.

HYDROGEN

99. They are contrasted with theories, because they may not yet be supported by evidence. In the scientific method, they come after the research question, but before experimentation. What are these predicted outcomes in a scientific study, often referred to as educated guesses?

HYPOTHESIS (Accept HYPOTHESES)

100. This is one of the stages of the water cycle. In this stage, water vapor in the air gets cold and changes back into liquid form. What is this stage of the water cycle, responsible for the formation of clouds?

CONDENSATION

101. This field of science includes cosmology, and is often considered a branch of physics. Individuals who study this field might use devices named after James Webb, Kepler, or Hubble. What is this branch of science that studies the planets, stars, and other objects in space?

ASTRONOMY

102. This two-word phrase refers to how heat is trapped in Earth's lower atmosphere. It brings forth an image of where plants are cultivated. What is this effect, a large contributor to global warming?

GREENHOUSE (EFFECT)

103. This reverses its direction many times a second at regular intervals. It can be "stepped up" and "stepped down" by transformers. What is this means of transmitting electricity over long distances?

ALTERNATING CURRENT

104. Roller coasters convert gravitational potential energy into this type of energy. It can be calculated as one-half times mass times speed squared. What is this energy of motion?

KINETIC (ENERGY)

105. Unlike boiling, which takes place throughout a liquid, this process just takes place on a liquid's surface. It can be considered the opposite of condensation, and is an example of a physical change. Identify this change of state, which involves the transformation from liquid to gas.

EVAPORATION

106. This type of structure is often composed of chitin [ki-tin]. It can be contrasted with an endoskeleton. What is this external support structure found in insects and other arthropods?

EXOSKELETON [eck-soh-SKEH-luh-tun]

107. This apparatus is mostly responsible for cellular transport. It can be thought of as a packaging center, and vesicles are often found near it. Name this organelle, named after an Italian scientist, that goes by several similar names.

GOLGI [GOAL-jee] (APPARATUS) (Accept: GOLGI BODY or GOLGI COMPLEX)

108. This term refers to a form of energy that can be transferred from one object to another. It can be transferred through processes like conduction and convection. What is the name for the amount of molecular energy present, sometimes described as the opposite of coldness?

HEAT

109. It is important to protect yourself against this danger. It can happen even on a cloudy day. What is sunscreen designed to protect against?

SUNBURN (Accept: ULTRAVIOLET (RAYS OR RADIATION)) (Accept: UV)

110. This structure of life can be described by examples such as blood, muscle, and fat. It can only be found in organisms that are multi-cellular, as it requires more than one cell to be made. Identify this structure, defined as a group of cells working together to perform a job.

TISSUE

111. An example of this simple machine can be found at the top of a flagpole. It comes in fixed and moveable varieties. What simple machine consists of a wheel and a rope?

PULLEY

112. This chronic disease involves faulty regulation of insulin. It comes in types one and two. Name this health condition, marked by problems with blood sugar, and often associated with obesity.

DIABETES (MELLITUS)

113. This organ is divided into lobes, such as the frontal and temporal. It connects to the spinal cord at its namesake stem. Name this organ, located in the skull, which allows us to think.

BRAIN(S)

114. This layer of the Earth makes up most of the asthenosphere. It contains convection currents, on which the plates of the crust float. Name this layer of the Earth, which surrounds the outer core.

MANTLE

115. In humans, this organ connects to the esophagus and intestines. It contains gastric acid. What is this organ that digests food?

STOMACH

116. This value is a counterpart to the heat index. It decreases as temperature goes down and air flow goes up. Identify this factor often reported in the winter, an adjustment to temperature for wind.

WINDCHILL (FACTOR)

117. This is the temperature below freezing at which moisture in the air will condense on an exposed surface. It depends on the relative humidity of the air. What is this temperature called that corresponds to the dew point below freezing?

FROST (POINT)

118. This part of some invertebrates is composed of an outer waxy, water-resistant layer. It does not grow, but must be molted and a new one secreted. What do we call this covering, the lack of which would leave the animal soft and vulnerable?

EXOSKELETON [ek-soh-SKEL-i-tn]

119. This type of motion depends upon the relation between two moving objects. It does not use the Earth as a reference point. What is this calculation of the motion of one object, with regard to another moving object?

RELATIVE (MOTION)

120. This region is located above the thermosphere and borders outer space. It gradually fades into the vacuum of space. Identify this outermost region of the Earth's atmosphere.

EXOSPHERE

121. This element, along with nickel, makes up the heaviest elements that stars can make through fusion. It has an atomic number of 26 and symbol Fe. What is this element, which is reinforced with carbon to form steel?

IRON

122. This rocky planet has an orbit of 88 days, but takes 59 days to complete a rotation. It has an incredibly hot side, with a temperature of about 750 degrees Fahrenheit, but its dark side has temperatures as low as minus 328 degrees Fahrenheit. What planet is named for the Roman messenger god, and is the closest planet to the sun?

MERCURY

123. This type of radiation was used to determine the structure of DNA. It has higher energy than UV rays, but lower than gamma rays. What are these types of electromagnetic waves, often used by dentists and doctors to image bones and teeth?

X-RAY(S)

124. This is the largest internal organ in the human body. It helps digestion, and also detoxifies harmful materials. What is this organ, particularly damaged by alcoholism?
- LIVER
125. This organelle is the location of mitosis, and houses the organelle where ribosomes are made. It holds the DNA of the cell. Name this membrane-bound, central area of eukaryotic cells.
- NUCLEUS
126. This designation applies to more than 90 percent of all animals. It includes insects, arthropods, and worms. What group contains animals that have no backbones?
- INVERTEBRATE (Accept: INVERTEBRATA)
127. This human body system allows us to take in sensory information from the environment. It includes the spinal cord and the body's neurons. What is this organ system, best known for containing the brain?
- NERVOUS (SYSTEM)
128. This state of the atmosphere occurs in the troposphere. It involves temperature, humidity, precipitation, air pressure, and wind. Name this state that occurs at a particular place during a short period of time.
- WEATHER
129. A treatment for cancer sometimes used with chemotherapy is named for this term. It involves the giving off of heat in the form of electromagnetic waves. Give this term, which can refer to a form of heat transfer that can occur in empty space?
- RADIATION
130. This rock is made up of quartz and feldspar. It is a coarse or medium-grained igneous intrusive rock. What is this type of igneous rock, often used in countertops and tombstones?
- GRANITE
131. This substance has the highest value on the Mohs hardness scale. It is a form of carbon that is created under very high pressure. Name this type of gem, one of the more expensive materials on Earth.
- DIAMOND
132. This type of appendage is usually associated with invertebrates. It is a long, slender, flexible organ that grows near the mouth, and lets the organism move, grasp objects, feed, and touch. Name this grabby appendage possessed by cuttlefish, squid, and octopus.
- TENTACLE(S)
133. An object that is described by this word can attract an electrically uncharged body. It is a property of the element iron, and its name comes from this fact. Starting with "F", what is the descriptive word that describes a substance in which the magnetic moments of the atoms are aligned?

FERROMAGNETIC

134. The rate of change in this quantity is known as jerk. It is a subject of Newton's second law, where it is multiplied by mass to give force. Give the term for the change in velocity over a given time, the rate at which something speeds up or slows down.

ACCELERATION

135. This process occurs in many desert organisms. It is a survival technique typically of a shorter duration than hibernation. Name this biological process, in which animals go dormant when it is hot and dry.

ESTIVATION

136. This process is used in sonar, and can happen to light or sound. It is a transformation of waves, but isn't diffraction or refraction. Name this phenomenon, which can be used to see yourself in a mirror.

REFLECTION

137. This is a scientific measurement. It is also an important measure that your doctor will take each time you visit him. What is this measure of the effect of gravity on an object?

WEIGHT

138. This layer of soil is full of minerals and organic matter, and is a good layer for plants to live. It is also known as the A horizon. Name the outermost layer of soil.

TOPSOIL

139. This phenomenon is leading to the melting of the polar ice caps. It is caused by the greenhouse effect and made worse by fossil fuels. Give the term used to refer to the recent increase in average temperatures throughout the Earth.

GLOBAL WARMING (Accept CLIMATE CHANGE)

140. At this temperature, condensation can occur. It is characterized by the presence of humidity and can raise the heat index. Name this two-word weather phrase, named for a substance that forms in the air once this temperature is reached.

DEW POINT

141. When sugar is dissolved into water, this type of mixture is formed. It is created any time one substance is dissolved in another. Identify the type of homogeneous mixture that can be separated.

SOLUTION

142. Special types of this process are called C4 and CAM. It is done by cyanobacteria and algae, and takes place in the chloroplasts. Name this process by which plants generate sugars from water and carbon dioxide.

PHOTOSYNTHESIS

143. The process of cold fusion could take place near this temperature. It is equal to negative 273 degrees Celsius, or zero Kelvin. What is this two-word theoretical temperature, when all molecular motion stops?

ABSOLUTE ZERO

144. Types of it include bituminous and anthracite. It is extracted through mountaintop removal, strip mining, and shaft mining. Name this fossil fuel, mined in much of Eastern Kentucky.

COAL

145. This structure in insects, crabs, and lobsters is made of chitin [KIGH-tun]. It is formed from the epidermis, and has tiny holes called spiracles [SPY-ruh-kuls] for breathing. What name is given to the outer skeleton on some arthropods?

EXOSKELETON

146. Currents in this layer of the Earth may be responsible for the planet's magnetic field. It is made of liquid nickel and iron, and lies below the mantle. Name this layer of the earth, the planet's second deepest layer.

OUTER CORE

147. This tree is the only living representative of its order. It is native to China, but is valued worldwide as an attractive fungus and insect resistant tree. What is this tree with fan-shaped leaves, and deeply furrowed grayish bark?

GINKGO [GING-koh]

148. The gravitational type of this type of energy is equal to gravity times mass times height. It is often contrasted with kinetic energy. What is this energy that an object has at rest?

POTENTIAL (ENERGY)

149. Each engine of a 747 airliner generates over 43,000 pounds of this force. It is one of the four forces that act on an airplane, and it counteracts drag. Name the force that propels an object in the direction of motion, especially when generated by the object itself, as by an engine or rocket.

THRUST

150. It has the densest atmosphere of any planet in our solar system, which is composed largely of carbon dioxide. It is one of two planets in the solar system with no known satellites. What is this planet, the closest to the Earth?

VENUS

151. This product of mass and velocity is sometimes represented by the letter P. It is related to the measure of kinetic energy. Give this term that is often used to refer to the amount of velocity an object has built up.

MOMENTUM

152. Some scientists think this structure may have been created when a large object struck the Earth. It is responsible for the rise and fall of the tides. Identify this only naturally occurring satellite of the Earth.

(THE) MOON (Accept: LUNA)

153. The structure of this molecule was discovered by Watson and Crick. It is said to have a double-helix shape. What is this molecule that holds the genetic code for life on Earth?

DNA (Accept DEOXYRIBONUCLEIC ACID)

154. Speed is equal to this value, divided by time. It is similar to displacement, and cannot be negative. What is this measure of how far an object travels?

DISTANCE

155. The "terminal" type of this quantity describes when a falling object reaches its fastest possible speed. It is similar to speed, but also includes a specific direction. Identify this property of matter, which measures an object's rate of change in position.

VELOCITY

156. In science experiments this should be written like a detailed recipe. It is so detailed that anyone should be able to duplicate the experiment exactly. What is this plan of how the experiment will be conducted?

PROCEDURE

157. This type of radiation has frequency between visible light and microwaves. It is sometimes used in sensors and remote controls. What is this type of radiation, with wavelength just longer than red light?

INFRARED (RADIATION) (Accept: INFRARED WAVES)

158. This is a very narrow band in the spectrum of electromagnetic radiation. It is surrounded by radiation that produces heat on one side, and nasty sunburns on the other. What do we call that part of the spectrum that can be detected by the human eye?

(VISIBLE) LIGHT

159. Bursts of this type of electromagnetic radiation have been observed in distant galaxies and are extremely bright. It is the electromagnetic radiation with the shortest wavelength and the highest energy. What is this radiation that can pierce through most objects other than several layers of lead [led]?

GAMMA (RAY)

160. This bone connects to the top of the tibia and fibula. It is the largest bone in the human body. What is the proper name for the thigh bone?

FEMUR

161. This planet is thought to be the reason Jupiter is not closer to the sun. It is the least dense planet in the solar system, and its largest moon is Titan. Name this second largest planet in the solar system.

SATURN

162. In humans, this organ can be separated into chambers known as the atria and ventricles. It is the main organ in our circulatory system. What is this organ that pumps blood through the body?

HEART

163. This property is measured from crest to crest or trough to trough. It is the opposite of frequency. What is this property of a wave, often measured in meters?

(WAVE)LENGTH

164. We use this term to name the sum total of all chemical reactions that take place within an organism's cells. It is the process by which your body converts what you eat and drink into energy. What is this process that is sometimes erroneously blamed for excessive weight gain?

(SLOW) METABOLISM

165. This number is named in honor of an Austrian physicist, psychologist and philosopher. It is the ratio of the speed of a projectile, in relation to the speed of sound in the same medium. What is the term for this number?

MACH (NUMBER)

166. During a total solar eclipse, you have a rare opportunity to see this outermost layer of the sun's atmosphere. It is the region of the sun where gases become the solar wind. Identify this "crowning" ring of the sun.

CORONA

167. When heavy rains occur, this layer can be eroded through runoff. It is typically less than 10 inches thick, and contains many minerals and micro-organisms. Identify this term for the upper layer of soil.

TOP(SOIL)

168. Absolute zero on this temperature scale is equal to negative 273 degrees. It is used in most countries throughout earth, but not often in the United States. What is this temperature scale, where water freezes at zero degrees?

CELSIUS (Accept: CENTIGRADE)

169. This plant structure enables common ornamentals like the tulip to produce flowers rapidly. It is usually globe-shaped and found underground. What is this structure that is the resting stage for some plants, and enables them to lie dormant when water is unavailable?

BULB

170. This type of medication is ineffective against viruses. It is, however, used to treat pneumonia, strep throat, and staph infections. What kind of substance, such as penicillin, is cultivated from microorganisms in order to destroy bacteria?

ANTIBIOTIC

171. This layer of the atmosphere is where most clouds are found. It lies between the Earth's surface and anywhere from six to twelve miles up. Name this layer where weather occurs.

TROPOSPHERE [TROP-uh-sfeer]

172. This layer of the atmosphere is just below the mesosphere. It lies just above the troposphere. Identify this layer of the atmosphere, which shares its name with a Las Vegas skyscraper and hotel.

STRATOSPHERE

173. This law of electrical resistance was established following investigations by a German physicist for whom it is named. It may be expressed in the equation resistance equals voltage divided by current or amperes. What is this law named for the German physicist?

OHM'S (LAW)

174. This form of heat transfer is typically contrasted with convection and radiation. It occurs on your stove top. What is this form of heat transfer, that relies upon direct contact?

CONDUCTION

175. This meteorological event occurs every year, once in the spring and once in autumn. It occurs when the sun shines directly on the equator. Name the meteorological event during which the lengths of the day and night are equal.

EQUINOX

176. This structure is an alternative to the placenta, in animals that lay eggs. It provides nourishment for growing embryos. Give this term for the part of an egg that is typically yellow in color.

YOLK

177. This force is overcome by thrust. It refers to the forces that act in opposition to an object moving in fluid. Identify this term, often associated with forces of air on an airplane.

DRAG

178. This process begins when light energy is captured by a plant. It removes carbon dioxide from the air and produces oxygen in its place. What is this process that changes light energy into chemical energy?

PHOTOSYNTHESIS [foh-tuh-SIN-thuh-sis]

179. Along with the brain, this structure's cord makes up the central nervous system. It separates vertebrates from invertebrates. What is this part of the skeleton, that is segmented and supports the torso?

SPINE (Accept BACKBONE)

180. This substance is made from fossilized tree resin. It sometimes preserves prehistoric insects or feathers. Name this material that is orange in color and often used in jewelry.

AMBER

181. This phylum is the largest in the animal world. It takes its name from Greek words that mean "jointed leg." What phylum contains millipedes, crabs, spiders, and insects?

ARTHROPOD (Accept: ARTHROPODA)

182. When this event occurs, the ocean and climate interact to cause drier than normal weather conditions in the Pacific Northwest and wetter than normal conditions in the Gulf Coast. It takes its name from Spanish for the "Christ child." What event causes warmer than average water temperatures in the Pacific Ocean during winter?

(EL) NIÑO

183. This temperature scale is standard in almost all countries that have adopted the metric system. It takes its name from the Swedish astronomer who invented it. Name this scale that uses one hundred degrees for the boiling point of water.

CELSIUS [SEL-see-uhs]

184. This plant process may be initiated by such things as light exposure, absorption of water, and warming. It usually follows a period of dormancy. What do we call the sprouting of a seed?

GERMINATION

185. This long-term process occurs through the process of natural selection. It was described in Darwin's *On the Origin of Species*. Name this process by which organisms change over time.

EVOLUTION

186. The presence of this element in paint can lead to poisoning. Its atomic number is 82 and its symbol is PB. Give this element, the namesake of a type of pencil that actually contains graphite.

LEAD

187. This laboratory instrument includes a collar to control the amount of air it receives, and a rubber tube to connect to a gas source. Its inner cone is a blue flame, while its cooler outer cone is more pale. What is this gas burner, named for the German chemist who invented it?

BUNSEN (BURNER)

188. This tree is sometimes called a "living fossil" and has an irregular branch pattern. Its leaves have long been used as a health supplement, and some claim it helps memory. What is this Asian tree, one of the oldest varieties still existing?

GINKGO (BILOBA)

189. Parts of this system include the mouth and the small intestines. Its main function is the breakdown and absorption of food. Identify this organ system, which includes the stomach.

DIGESTIVE (SYSTEM)

190. This invertebrate generally has five hollow arms covered with short spines. Its rows of tube feet allow it to creep in any direction. Name this marine organism whose arms or rays surround a central disk.

SEA STAR (Accept: STARFISH)

191. Messages on this site are restricted to 140 characters. Its symbol is a characteristic blue bird. Identify this social media site, where people can follow, like, and "tweet."

TWITTER

192. It makes up the majority of the integumentary [in-TEG-u-men-TARE-ee] system. Its top layer is known as the epidermis, and UV light can cause it to burn. Name this outer layer of cells that protects humans, and gives rise to hair and nails.

SKIN

193. These are important scientific tools used in most biology labs. Light and electron are two types of these scientific tools. What are these important laboratory tools?

MICROSCOPE(S)

194. Scientists in this field study phenomena that occur in the troposphere and lower stratosphere. Like climatology, it examines atmospheric processes. Identify this science, the basis of weather forecasting.

METEOROLOGY

195. When these areas switch for Earth, it is called a geomagnetic reversal. Like ones of these for two magnets repel each other, while unlike ones attract each other. What ends of a magnet are commonly called north and south?

POLES

196. The largest of these is a dwarf planet named Ceres. Many can be found in the scattered disk. What are these objects in space, which give their name to a belt between Mars and Jupiter?

ASTEROID(S)

197. There are more known examples of animals from this phylum than any other. Members include the extinct trilobites, and they all have exoskeletons and segmented bodies. What are these animals, which include crabs, spiders, and bees?

ARTHROPOD(S) (Accept: ARTHROPODA)

198. Recent evidence suggests that dinosaurs might be classified this way. Modern animals classified this way include birds and mammals, which are able to regulate their body temperature internally. What type of animal group fits this description?

WARM BLOODED (ANIMAL) (Accept: ENDOTHERMS or ENDOTHERMIC)

199. It is the SI unit of electromotive force. Most of the electrical outlets in the U.S. carry about 120 of them. What is this unit of electricity, which is used along with amps, watts, and ohms?

VOLT(S)

200. This type of relationship is seen with mistletoe, although it can do photosynthesis. Most of the time, though, it gets its food from a host. In what kind of relationship does one organism benefit, while the other is harmed?

PARASITIC (Accept other forms of the word)

201. This specific bacterium normally lives in the intestines of both people and animals. Most varieties are harmless to humans, but some strains can cause people to get sick from contaminated food and water. Name this bacterium that was linked to an outbreak of illnesses at the Chipotle Mexican Grill chain of restaurants.

E.COLI

202. Nobel Prize winners in this branch of science include Ernest Rutherford, Linus Pauling, and Marie Curie. Name the branch of science that studies the composition of matter, and how different elements bond to make molecules.

CHEMISTRY

203. Examples of this climate can be found in Paraguay and Kentucky. Name the climate that consists of seasons without extremes in precipitation or temperature.

TEMPERATE (CLIMATE)

204. This form of energy can be seen in Van de Graaff generators and lightning. Name the energy of electrons moving from place to place, that comes in varieties like current and static.

ELECTRICITY (Accept other forms of the word)

205. Discovered by Huygens [HIGH-gunz] in 1655, this moon is the only one in the solar system with clouds and a dense atmosphere. Name the largest of Saturn's moons.

TITAN

206. These objects can attract or repel certain kinds of metal and have two poles. Name the objects that have a north and south pole and generate a namesake field.

MAGNETS

207. These cave formations may form from lava or limestone, and their name comes from their formation through dripping. Name the rock-like structures that hang from the tops of caves.

STALACTITES

208. This scientist observed plants and cork under a microscope, leading to a novel discovery. Name the scientist who, after observing the honeycomb-like shape of plant parts under a microscope, coined the term "cell."

(ROBERT) HOOKE

209. These units are made up of a core of neutrons and protons and a cloud of electrons, and are considered the building blocks of matter. Name the smallest units into which matter can be broken up and still remain the same element.

ATOMS

210. Two types of these structures can be found on conifers. One produces pollen, and the other contains the ovule. Name the structures found on coniferous trees that contain the cells needed for reproduction.

CONES

211. These devices are used to either increase or decrease the strength of electricity. Name these devices that can often be found in cylinders on utility poles.

TRANSFORMER(S)

212. These have many chloroplasts and are, therefore, the primary place photosynthesis occurs. Name these structures of plants, that fall off of many types of trees during autumn.

LEAF (Accept: LEAVES)

213. This body part in animals often consists of spines or bone covered by webbing. Name this body part, examples of which include pectoral and dorsal, and which allow fish to swim.

FIN

214. This process can involve the change of a particular material from sedimentary to metamorphic, or vice-versa. Name this cycle that involves the transformation of solid materials.

ROCK (CYCLE)

215. This lab device is named for a scientist who heated up elements to see what colors they gave off when burned. Name this device used to produce a flame, found in many science laboratories, and named after a German chemist.

BUNSEN BURNER

216. The foam version of this device was invented by Aleksandr Loran in 1904, and more recent versions have used sodium bicarbonate and Halon to cut off oxygen. Name this device, often holding pressurized carbon dioxide or nitrogen, used in emergencies to put out fires.

(FIRE) EXTINGUISHER

217. This scientist was an assistant of Tycho Brahe [TEE-ko BRA-hee]. Name this German astronomer, known for his laws of planetary motion.

(JOHANNAS) KEPLER

218. This area of the body is where an insect's legs and wings are attached to the body. Name this insect body part that is between the head and the abdomen.

THORAX

219. This form is generally very different from the adult form and is common in many insects. Name this life cycle stage that occurs after hatching for insects that undergo complete metamorphosis.

LARVA (ACCEPT: LARVAE)

220. This layer of the atmosphere is where all weather occurs. Name this lowest layer of the atmosphere.

TROPOSPHERE

221. This animal, found in Central and South America, lives in burrows that it digs into the ground and feeds on termites and other insects. Name this mammal who can use a set of plates along its back as protection, in some cases forming a solid ball.

ARMADILLO

222. This rock is made up of recrystallized carbonate minerals, and has a unique textured appearance. Name this metamorphic rock, often used in countertops, columns, and sculptures.

MARBLE

223. This term refers to the movement of sediment and earth by wind, water, or other natural forces. Name this natural process, which is related to but contrasted with weathering.

EROSION

224. This part of human anatomy is often 3 to 4 inches long, but its purpose has been unclear. Some scientists now think it stores beneficial bacteria in case of emergency. Name this part of the intestine, the inflammation of which is dangerous, and often requires surgery.

APPENDIX

225. One of its best known features is the great dark spot. Name this planet, the farthest from the sun.

NEPTUNE

226. During this time, an animal's metabolism slows down greatly and relies largely on stored fat. Name this process, commonly seen in bears, during which the animal is largely inactive during the winter.

HIBERNATION

227. This season includes the first equinox of the year and sees widespread plant pollination. Name this season that occurs between winter and summer.

SPRING

228. It is associated with the reversal of wind patterns and the asymmetrical heating of land and sea. Name this seasonal climate pattern of South Asia and the Indian Ocean, marked by heavy rainfall.

MONSOON

229. This derivative of glucose has a structure similar to cellulose, and is pliable but very tough. Name this substance that makes up the exoskeletons of arthropods and crustaceans [krust-EH-shuns].

CHITIN

230. It is bound by the cell membrane and includes various cellular organelles. Name this substance, that contains the contents of the cell that are not within the nucleus.

CYTOPLASM [SIGH-toe-pla-zum]

231. Laws by Kepler help explain why these are shaped like ellipses. Name this term that refers to the path a planet takes when revolving around the sun.

ORBIT

232. These vascular plants have roots, stems, and leaves. Name this type of plant that bears no flowers, and reproduces not by seeds or pollen, but by spores.

FERN(S)

233. This memory device is organized according to decreasing wavelength. Newton originally only considered five of the pieces of this acronym, before adding orange and indigo. Give the acronym for the hues of human vision that make up a rainbow.

ROY G. BIV (Accept: "RED-ORANGE-YELLOW-GREEN-BLUE-INDIGO-VIOLET")

234. This scale was discovered in 1724 by the Polish-Dutch physicist for whom it was named. Normal body temperature is approximately 98.6 degrees using it. What temperature scale has water freezing at 32 degrees and boiling at 212?

FAHRENHEIT

235. The incomplete type of this process includes a nymph stage. Notably, butterflies and frogs undergo this process. What is this life cycle process, which involves transitions between phases such as larva and pupa [PEW-pa]?

METAMORPHOSIS [met-uh-MAWR-fuh-sis]

236. This object is orbited by several dwarf planets, many asteroids, and eight planets. Nuclear fusion occurs here, leading to production of light. Identify the star at the center of our solar system.

(THE) SUN

237. This term refers to speeds that are higher than Mach [mock] one. Objects with this type of speed often create a namesake boom, and there was once an NBA team in Seattle with this term as its nickname. What is this term for objects that travel faster than the speed of sound?

SUPERSONIC

238. Temperatures on this scale can be converted to Kelvin by subtracting 32, then multiplying by five-ninths, and then adding 273. On this temperature scale, the boiling point is 212. Name this temperature scale, the most popular one in the United States.

FAHRENHEIT (TEMPERATURE) (SCALE)

239. This is the second most abundant mineral in the Earth's continental crust. One gem that is made from this mineral is an amethyst. What mineral has been used in watchmaking for the past 40 years?

QUARTZ

240. They are the only two planets in our solar system without moons. One is the smallest planet, and the other is the second-brightest object in the night sky. Identify these two innermost planets of our solar system.

MERCURY (AND) VENUS (either order)

241. This scientist was originally an assistant to Tycho Brahe [BRAH-hay], who inspired much of his work. One of his laws states that planets have elliptical orbits. Name this astronomer known for his laws of planetary motion.

(JOHANNES) KEPLER

242. This man is known for having feuds with Nikola Tesla. One of his most famous inventions was the phonograph. Name this inventor, known best for his light bulbs.

(THOMAS) EDISON

243. These two qualities of measurements are important in science. One of the qualities is described as exactness of the measurement and the other is the ability to repeat a measurement with similar results. What are these qualities of measurements?

ACCURACY, PRECISION

244. These arthropods include wood lice, but most of them are aquatic. One variety has legs that can span up to 12 feet in diameter. What kind of arthropods include shrimp, crabs, and lobsters?

CRUSTACEANS (Accept: CRUSTACEA)

245. These substances are chemical messengers produced by the endocrine system. Ones that are associated with stress include cortisol and adrenaline. What are these substances, which include estrogen and testosterone?

HORMONE(S)

246. Scientists in this discipline might study angiosperms and gymnosperms. Organisms they study typically use photosynthesis for energy. Who are these scientists that study plant life?

BOTANISTS

247. Molecules regulated by this system include Human Growth Hormone, cortisol, and testosterone. Organs of this system include the pituitary, pancreas, and other glands. What is this organ system, which controls hormones in humans?

ENDOCRINE (SYSTEM)

248. They are thought to be the oldest domain of life on Earth. Originally, they were thought to be types of bacteria. Identify these organisms, unicellular lifeforms that sometimes live in extreme environments.

ARCHAEA [are-KAY-uh]

249. Examples of these changes known as changes of state include evaporation, condensation, and freezing. Other examples including cutting material or creating mixtures. What are these changes that do not change the chemical makeup of a substance?

PHYSICAL (CHANGE)

250. One of the largest orders in this class consists of the flying members, the bats. Others in this class include the egg-laying monotremes. What is this class of animals, examples of which include echidnas, whales, and humans?

MAMMAL(S) (Accept: MAMMALIA)

251. In space, the types of these include elliptical, spiral, and irregular. Our solar system is part of a spiral one, named the Milky Way. What are these that are defined as a system of stars, dust, and dark matter that are bound together by gravity?

GALAXY **

252. This structure can be made of bone or cartilage, and is found in vertebrates. Parts of it can include human structures such as the femur. What is this type of internal support system, contrasted with the types of skeletons found in many insects?

ENDOSKELETON

253. This cycle is also known as the hydrologic cycle. Phases in this cycle include condensation and evaporation. Name this cycle by which H₂O is naturally filtered and distributed.

WATER (CYCLE)

254. Thanks to widespread vaccination, today this disease is only found in Afghanistan and Pakistan. Regular and oral vaccines were developed in 1955 and 1961 by Jonas Salk and Albert Sabin, respectively. What is this infectious virus that attacks the brain and spinal cord, sometimes causing paralysis in young children?

POLIO(MYELITIS)

255. Its layers include the lithosphere, asthenosphere, mesosphere, and atmosphere. Relative to its size, it has the largest satellite in the solar system. What is this planet we call home?

(THE) EARTH

256. Many animals with this type of temperature regulation will spend time basking in the sun. Reptiles and amphibians are this type of animal. Give this term for animals that do not regulate their own temperature.

COLD-BLOODED (Accept: ECTOTHERM(IC))

257. It can be considered a measure of average heat over a given space. Scales to measure it include the Kelvin and Celsius scales. What is this property, often measured with a thermometer?

TEMPERATURE

258. Examples of animals in this order include aye-ayes [eye-eyes], a type of lemur. Several member species of this group walk on two feet, including the various gibbons. What is this grouping of animals, which includes gorillas, chimpanzees, and humans?

PRIMATE(S)

259. Jane Goodall is famous for her work with these primates. She studied these animals for 55 years in Tanzania. What are these animals that can use tools?

CHIMPANZEES (Accept: CHIMP)

260. The term "perigee" refers to the point of the Moon's orbit when it's nearest to the Earth. Similarly, this term is the point at which the orbiting object is closest to the Sun. Name the term that is given to a planet's or comet's point in orbit, where it is closer to the Sun than any other point.

PERIHELION

261. Edmond Halley created the world's first weather map, and showed these winds on them. So named by business-minded mariners, these winds are created by the deflection of warm air flows by the rotation of the Earth. Meeting at the doldrums near the equator, identify these winds that blow from the northeast and southeast.

TRADE (WINDS)

262. These phenomena are more frequent than rainbows. Some are called upside down rainbows, and others are called sundogs. What are these shapes, created by the refracting and reflecting of light by ice crystals in the sky?

(ICE) HALOS

263. This index is a ratio of weight to height squared. Some argue that body fat percentage is a better marker of health than this value. What is this health index, which is used to diagnose obesity?

BMI (Accept BODY MASS INDEX)

264. Animals of this type often have bacteria in their stomach to digest cellulose. Some of them have large molars, while insects of this type often feed on leaf tissue. What are these animals that eat primarily plants?

HERBIVORE(S)

265. This part of the seed develops from what it is called an integument [in-TEG-yuh-munt]. Sometimes it sticks to the seedling leaves. What part of a plant seed protects the seed?

SEED COAT

266. These scientists can be seen during news reports on television. Sometimes they are called weathermen. What scientists study the weather and climate?

METEOROLOGIST(S)

267. In 2015 the National Science Foundation awarded a grant to researchers at MIT to develop this specific technology. Specifically developed robots will be able to do this for children. What job will these robots do to help preschoolers learn an important skill?

READ

268. Used in mineralogy, this scale typically ranges in value from one to ten. Students typically learn that talc is found at the low end, and diamond at the high end. Give the name for this scale of mineral hardness.

MOHS

269. This term identifies species of flowering plants, the largest group in the plant kingdom. Such a flowering plant contains an ovary within its flower where the seed develops. Identify this type of plant which represents about 80 percent of all known green plants.

ANGIOSPERM [AN-jee-uh-spurm]

270. These worms secrete a substance called hirudin [HIR-yuh-din], which keeps blood from clotting. That is why they were once used by doctors to drain off blood. Identify these blood-sucking worms that are often found in swampy areas.

LEECHES

271. This substance is transferred from the anther to the stigma. That movement of this substance can involve bees or other insects. Give the term for this reproductive substance used by many plants.

POLLEN

272. These animals are members of the order Diptera. The "fruit" version of these are often used in science experiments. Name this type of winged insects.

FLIES (Accept FLY)

273. The smallest variety of this aquatic invertebrate is only 2 inches long. The blue-ringed type is one of the most venomous sea creatures known. Identify this sea creature that has a name that means "eight legs."

OCTOPUS

274. The frozen component of this sphere of the Earth is known as the cryosphere [CRY-uh-sfeer]. The components of this sphere go through a cycle that contains the steps evaporation and condensation. Which sphere of the Earth contains all of Earth's oceans, lakes, rivers, and other bodies of water?

HYDROSPHERE

275. This weather phenomenon often occurs on surfaces in the winter. The first of these in a given year can kill plants. What is this term for crystallized condensation on surfaces, that happens when water comes in contact with a cold object?

FROST

276. Insects have an "exo" one, made of chitin [KIE-ten]. The human one is made of bones and connective tissues. What is this organ system, that provides structure for organisms?

SKELETON (Accept: SKELETAL SYSTEM or SKELETON)

277. Most of these fish are extinct. The lamprey is an example of these types of fish. What kind of fish are named for their missing body parts?

JAWLESS (FISH)

278. As opposed to normal larvae, this stage of organism typically resembles the adult form. The name of this life stage comes from nature spirits in Greek mythology. What is this middle stage of incomplete metamorphosis?

NYMPH

279. The name of Lava soap was inspired by this, one of its key ingredients. The name of this rock means "foam" in Latin, likely because of it is full of little air pockets. Able to float, name the porous rock that is solidified lava.

PUMICE

280. This part of the plant is encased in an ovary. The ovary is often called a fruit. What part of a plant do farmers use to produce their gardens?

SEED

281. If these substances are heated further, they could lead to the formation of a plasma. The right-most column of the periodic table is known as the "noble" ones of these. What is this state of matter, in which hydrogen and helium naturally exist on Earth?

GAS(ES)

282. Bacteria can exist as one of these types of organisms. The root word of this term means "one" or "single". Identity the term for an organism that is composed of a single cell.

UNI-CELLULAR

283. Types of this process include conjugation, budding, and binary fission. The sexual type of this process involves cells known as gametes. What is this process by which organisms produce offspring?

REPRODUCTION

284. These particles are much smaller in mass than particles contained in the nucleus. Their area of the atom is often referred to as orbitals or clouds. Name these negatively charged subatomic particles.

ELECTRON(S)

285. Members of this plant group have complex vascular bundles and fibrous root systems. Their floral parts usually occur in multiples of three, and their veins run parallel. Identify this class of angiosperms, whose name comes from the fact that their embryos contain only one seed leaf.

MONOCOT(YLEDON)

286. These substances can take on the shape of their container. Their molecular energy is between that found in gases and solids. What is this phase of matter, represented by room temperature water?

LIQUID(S)

287. These objects are often metals and allow electricity to move freely. There are also objects similar to these of the "super" and "semi" varieties. What is this type of object, which is contrasted with an insulator?

CONDUCTOR(S)

288. This moon has an atmosphere that is mostly oxygen, with traces of water and hydrogen. There are suggestions that a liquid ocean may lie beneath its icy crust. Name this moon that orbits the solar system's largest planet.

EUROPA [yoo-ROH-puh] (Accept: JUPITER II)

289. This kind of cell can divide to produce more of its own kind, plus other specialized kinds. There are two major kinds: embryonic and adult. What kind of cell may have the potential to replace defective or damaged cells within the body?

STEM (CELLS)

290. Many developing countries lack this important resource. There must be enough of this resource for everyone to be healthy. For what resource will cities issue boil advisories, if it is not safe to drink?

(SAFE) WATER

291. As the most advanced organisms on Earth, these animals include man. These animals are usually larger than animals without backbones. What type of animals have backbones?

VERTEBRATES

292. One of the steps in identifying these materials determines the color of powder produced, and is called a streak test. These materials' other properties include cleavage and hardness. What naturally occurring substances have a crystalline structure, examples of which include gypsum and quartz?

MINERALS

293. A nova sometimes occurs as this object's life cycle ends. These objects produce large amounts of energy through nuclear fusion. Name this astronomical object, examples of which include Alpha Centauri and the Sun.

STAR(S)

294. These phenomena that occur on the sun can have effects that can directly affect the Earth. These occur when magnetic energy on the sun has built up in the solar atmosphere, and is suddenly released. What are these occurrences on the sun, that are seen as a sudden, rapid, and intense variation in brightness?

(SOLAR) FLARES

295. These parts of a plant are its reproductive parts. These plant parts may be large or small, and a variety of colors. What plant parts bloom at different times of the year?

FLOWERS

296. This biome is one of Earth's coldest and harshest. These regions are treeless, where the climate is cold and there is little rainfall. What is this biome, found in the Arctic and on top of mountains?

TUNDRA

297. Two of this scientist's research subjects were nicknamed Fifi and David Greybeard. These research subjects were chimpanzees living in Tanzania. Name this anthropologist known for her work on nonhuman primates.

(JANE) GOODALL

298. These organs are part of the urinary system. These two organs are located in the lower part of the abdomen. What two organs filter blood and create urine?

KIDNEY(S)

299. These animals communicate using a unique waggle dance. They are a eusocial species, led by a single reproducing queen. What are these insects, known for their important role in pollinating flowers?

(HONEY)BEES

300. These substances speed up chemical reactions. They are all proteins, made of amino acids. Give the name of these natural catalysts.

ENZYMES

301. An example of these substances includes collagen. They are composed of amino acids. Identify this type of molecule, which is essential in human diets, and found in meats, nuts, and legumes.

PROTEIN

302. They are one of the three types of blood vessels. They are connected to veins by the capillaries. Identify these blood vessels that carry oxygen-rich blood away from the heart.

ARTERIES

303. Examples of these for carbon include its 12 and 14 forms. They are different than isomers, despite the similar name. What are these variations on an element, with different atomic weights?

ISOTOPES

304. Nephology, a sub-branch of meteorology, refers to the study of these objects. They are formed by condensation and they are involved in precipitation. Identify this weather phenomenon, types of which include cumulus, cirrus, and nimbus.

CLOUD(S)

305. Latin, scientific name, please. This class of animals is thought to have descended from the theropod dinosaurs. They are known for a unique mode of transportation, shared by insects and bats. Identify this group of animals, which includes the kiwis, emus, and penguins.

AVES [AY-veez]

306. The largest of these animals are the capybara. They are known for large incisors, and other examples include beavers and squirrels. What is this order of mammals, which include mice and rats?

RODENT(S) (Accept RODENTIA)

307. Recent evidence from the fossil record suggests that dinosaurs perhaps belonged to this group. They are known for their eggs and feathers. What is this phylum of organisms, which include kiwis, canaries, and penguins?

BIRD(S) Accept: AVES [ah-VAYS]

308. These parts of corn plants produce pollen. They are located at the top of the corn cobs. What are these silky looking strings?

(CORN)TASSELS

309. One of these devices is named for Foucault [fue-KOE]. They are made of a rod and weight, which oscillate back and forth. Identify this type of device, that swings back and forth.

PENDULUM

310. These members of the class Anthozoa form colonies from their individual polyps. They are members of the phylum Cnidaria [nye-DARE-ee-uh]. What are these colorful animals that often form reefs?

CORAL(S)

311. Examples of these bodies include Triton and Titan. They are natural satellites that orbit planets. What are these astronomical bodies, one of which causes the tides on Earth?

MOON(S)

312. These invertebrates were long thought to be plants. They are now classed in the phylum Porifera [puh-RIF-uh-ruh]. The skeletons of aquatic animals are used in the bath?

SPONGES

313. These sources of energy come from decomposed plant and animal remains. They are often contrasted with "green" sources of energy. Name this type of energy source, examples of which include coal and natural gas.

FOSSIL FUEL(S)

314. This type of object does not allow for the free flow of electrons. They are often contrasted with conductors. What is this term referring to objects that do not easily transfer heat or electricity?

INSULATOR(S)

315. These plants come in summer and winter types, and examples include crabgrass, corn, and peas. They are often contrasted with perennials. What are these plants that complete their life cycle in a single year?

ANNUAL(S)

316. These storms are measured using the Saffir-Simpson Scale. They are often named, and examples have included Andrew and Katrina. What are these tropical storms, which originate in the Atlantic?

HURRICANE(S)

317. Technically, this word refers to any flowering plant whose above-ground stem doesn't become woody. They are often prized for their scent, taste, and medicinal values. Give the term for spices such as parsley, sage, and oregano.

HERB(S)

318. These organisms are typically non-motile and autotrophic. They are often separated into divisions, and include classifications such as angiosperms and monocots. Name these organisms, which use chlorophyll to facilitate photosynthesis.

PLANTS

319. These events are measured using the Fujita scale. They are particularly common in the central United States and in an area called Dixie Alley. Name these meteorological events that are caused by a rapidly rotating column of air and that begin as a funnel.

TORNADOES (Accept: TWISTERS)

320. These organelles may be a modified version of cyanobacteria. They are the location of photosynthesis and are found in most plant cells. What are these organelles, which are green in color and contain a similarly named pigment?

CHLOROPLAST(S)

321. They are also known as spiny anteaters. They are the only mammals other than platypuses to lay eggs. What are these animals, native to Australia?

ECHIDNA [ih-KID-nuh]

322. Fronds are examples of these plant structures. They are the primary location of photosynthesis. Identify these plant parts that are typically green and attached to stems.

LEAF (LEAVES)

323. These organs contain the alveoli [al-vee-OH-lee]. They are the primary organs of the respiratory system. Identify these organs that lie within the chest, and allow us to breathe.

LUNGS

324. These organisms are also known as pollywogs. They are the water-dwelling larval stages of most amphibians. What is this organism, an early life stage of frogs and toads?

TADPOLE

325. They belong to the filter-feeding phylum of Porifera. They are thought of as the simplest animals, unable to move or think. What are these animals, sometimes used as a household cleaning tool?

SPONGE

326. Sagittarius A-star is the location of a super massive one, at the center of our galaxy. They are thought to be surrounded by a disk of matter and light. What are these high mass objects in space, from which not even light can escape?

BLACK HOLE(S)

327. Examples of these organisms include blue-green algae and E. coli. They are unicellular and prokaryotic. Identify these organisms, some of which are present in yogurt, and others of which cause strep throat.

BACTERIA

328. These animals are perhaps best known for their medusa phase of life. They belong to the phylum Cnidaria [NYE-dar-ee-uh]. What are these sea creatures, known for their painful stings?

JELLYFISH

329. These weather disasters are characterized by a buildup of water in fewer than six hours. They can be caused by rainstorms, or by melting ice or snow. What are these types of rapid high water events, which can be the focus of a type of severe weather warning?

FLASH FLOOD(S)

330. Virgin Galactic hopes to become the first commercial group to provide public access to these vehicles. They can be used to propel nuclear weapons, and were vital to missions such as Gemini and Apollo. What are these vehicles, commonly used to transport rovers, raw materials, or humans into space?

ROCKET(S)

331. They are smaller than asteroids, but larger than normal space dust, and they may or may not strike the earth. They can become shooting stars. What are these objects, which may eventually become a meteorite?

METEOROID (Not METEOR)

332. Two word answer please. This is one of the largest known invertebrates, along with its "colossal" relative. They can measure over 40 feet long, and their only predators include large whales. Identify these large sea creatures, mollusks with eight arms, two tentacles and a bird-like beak.

GIANT SQUID

333. These cellular projections can be used for movement or taking in food. They can often be found on amoebae [uh-mee-bay]. Give this term, literally meaning "false foot."

PSEUDOPOD [SOO-duh-pod] (ACCEPT: PSEUDOPODIA)

334. An object that passes Mach [mock] 1 produces one of these. They can often be heard when rockets take off. Identify this very loud sound, which is created when the sound barrier is broken.

SONIC BOOM

335. They are the smallest parts that a compound can be broken into. They cannot be broken apart by a physical change. Identify this term that names a collection of bound atoms.

MOLECULE(S) (ACCEPT: FORMULA UNIT)

336. Types of these cells include monocytes and macrophages. They combine with plasma and red blood cells to make up the blood. Name these blood cells, involved in immune function.

LEUKOCYTE [LOO-kuh-sight] (Accept: WHITE (BLOOD CELLS))

337. These organisms can reproduce by budding, or by expelling reproductive cells through their mouths to be fertilized in the water. They come in black, blue, thorny, and horny varieties, and form polyps when mature. What are these marine invertebrates, whose stony skeletons form reefs?

CORAL

338. They contain organelles and are surrounded by a membrane. They come in prokaryotic and eukaryotic types, based on whether they have a nucleus. Identify these basic units of life.

CELL(S)

339. Galileo used one of these to discover Jupiter's moons Io and Europa. They come in various types, like refracting and infrared. What are these devices, used to see far away objects?

TELESCOPE

340. These X-shaped structures are normally found in the nucleus of eukaryotic cells. They contain segments of hereditary information called genes, and they are passed from generation to generation through reproduction. Name these structures, composed of DNA, that carry the traits of an organism.

CHROMOSOME

341. Examples of these include trunks and thorns. They contain vascular tissues such as the phloem and xylem. Identify this central stalk of a plant, where leaves and flowers are attached.

STEM(S)

342. The more than 1,000 species of these animals make up the class Asteroidea. They have radial symmetry, and are one of the best known members of the echinoderms [ih-KIE-no-derms]. What are these sea creatures, that can often be found on beaches, and can regenerate their five limbs?

SEA STAR(S) (Accept: STAR FISH(ES))

343. This invertebrate of the insect family has a parasitic relationship with humans. They hide in the daytime, and come out at night to feed. What is this reddish brown insect that loves living in mattresses?

BEDBUG

344. These organelles are created in the nucleolus. They line the rough E-R and are the location of translation. What are these organelles, where proteins are made?

RIBOSOME

345. These trees may be either needle-leaved or broad-leaved. They may be found in rainforests, high on mountain sides, or in the far north. What kind of trees keep their leaves through a year and into the next?

EVERGREENS (Not: CONIFERS)

346. These scientists might work with geographers, but focus on physical aspects of the land. They may study sedimentary and metamorphic substances. Identify these scientists, who study rocks and other components that make up the earth.

GEOLOGISTS

347. These are the primary explanation for the process of continental drift. They meet at faults and come in both oceanic and continental types. What are these sections of the Earth's crust?

(TECTONIC) PLATE(S) (Accept: PLATE TECTONICS)

348. Subtypes of these scientists include mycologists, botanists, and zoologists. They might study cells, ecosystems, or evolution. Who are these scientists that study life?

BIOLOGIST(S)

349. Specific types of these scientists include social, cognitive, and personality. They might study topics such as memory, twins, and stereotypes. Identify these scientists who are sometimes also healthcare providers, known for providing therapy.

PSYCHOLOGIST(S)

350. Most spiders have six of these organs, though some only have four or two. They move independently and in concert to build webs. Usually located on the underside of a spider's abdomen, to the rear, name these silk-spinning organs.

SPINNERET(S)

351. Although these agents can replicate and can manipulate their surroundings to ensure their survival, many do not consider them to be living. They require a host cell and do not have their own nuclei. What infectious agents often reproduce by killing their host cells, and produce Ebola and influenza?

VIRUSES

352. If you fail to use these, you may end up having to use an eye wash. They shouldn't be substituted with normal glasses. Name this type of protective eyewear, often used during chemistry experiments.

(SAFETY) GOGGLE(S)

353. These scientists are a type of biologist. They study things like ferns, flowers or corn. What are these scientists that specialize in plant study?

BOTANIST(S)

354. These parts of plants usually have a lamina attached to the plant by a petiole [peh-TEE-ohl]. They take in carbon dioxide from the air. Which plant parts use light energy to change carbon dioxide and water into sugar?

LEAVES

355. These plant parts come in fibrous and tap varieties. They take in nutrients and water from the ground. What are these plant parts, often eaten in the form of radishes and carrots?

ROOT(S)

356. These structures are colorful features found in angiosperm plants. They usually consist of a collection of petals and give rise to fruit. What are these plant parts, examples of which include roses and lilies?

FLOWER(S)

357. These animals roamed the Great Plains and were hunted close to extinction in the 1800s for their fur and meat. They were a staple in many Native American cultures. Name this large bovine animal, native to the Americas.

BISON (Accept: BUFFALO)

358. Materials that have this property include glass, water, and plastic wrap. This is an adjective that describes material that allows objects behind it to be seen clearly. What is this term, the opposite of opaque?

TRANSPARENT

359. Examples of this include clay, silt, and loam. This is defined as the layer of Earth in which plants grow. What is this black or dark brown material, typically consisting of a mixture of organic remains, clay, and rock particles?

SOIL

360. This is the layer of Earth's atmosphere in which the space station orbits. This layer of the atmosphere is directly above the mesosphere, and below the exosphere. What is this fourth, and largest, layer of the atmosphere?

THERMOSPHERE

361. These vertebrates are special kinds of mammals. This mammal has a mouth and feet that resemble a duck. What is this mammal that lays eggs?

(DUCK BILLED) PLATYPUS

362. This physicist and mathematician was born in the seventeenth century. This man was instrumental in the scientific revolution. Who is this English scientist, who has three laws that contain his name?

(ISAAC) NEWTON

363. He was an aerospace engineer and test pilot. This Ohioan is famous for his trips to space. What man was the first American astronaut to walk on the moon?

(NEIL ALDEN) ARMSTRONG

364. The moons of this planet are Phobos and Deimos. This planet is the second smallest in the solar system, after Mercury. What is this planet, often referred to as the Red Planet?

MARS

365. This planet is considered a gas giant, and was named by the Romans after one of their gods. This planet's turbulent atmosphere is evidenced by a giant storm, that has been known to exist since the seventeenth century-the Great Red Spot. What is this planet, the largest of the planets in our solar system, and the fifth planet from the sun?

JUPITER

366. This process can be broken into cata- and ana- types, which break down and build up substances. This process deals with how organisms process food taken in from the environment. What is this term, which is often described as being "high" or "low" when discussing weight loss or gain?

METABOLISM

367. For water, this property is equal to 1 gram per cubic centimeter. This property explains why a cubic inch of lead is heavier than a cubic inch of styrofoam. What is this property, defined as mass per unit volume?

DENSITY

368. This term from science is often used differently in non-scientific language to mean a guess or hunch. This scientific term is an in-between step between a hypothesis and a law. What is this scientific term, examples of which include one of gravity and one of evolution?

THEORY

369. First and last name answer. This scientist is one of four to win two Nobel Prizes, one for physics and another for chemistry. This scientist discovered the elements Radium and Polonium. Who is this scientist, the French-Polish spouse of fellow scientist Pierre?

MARIE CURIE

370. In North America, it is during this season that the sun is closest to the Earth. This season is caused due to the tilt of the Earth, with the Earth being tilted away from the sun. What is this season, when days are shortest and nights are longest?

WINTER

371. The Breakthrough Starshot program might use very strong examples of these to power small spacecrafts. This term is an acronym standing for light amplification by stimulated emission of radiation. Give this term for a beam of focused light, sometimes used in namesake pointers.

LASER(S)

372. Words to describe it include tropical and temperate. This term is similar to weather but different in timescale. Identify this term for the long-term temperature and precipitation patterns of an environment.

CLIMATE

373. Formal name, please. This type of cloud is capable of producing thunderstorms, and other dangerous severe weather. This type of cloud is also capable of producing tornadoes. What cloud type has the shape of an anvil, and is considered a thunderstorm cloud?

CUMULONIMBUS (CLOUDS)

374. Two-word answer. Its first part, known as the duodenum, connects to the stomach. This winding organ is nearly six meters long when stretched out. Identify this organ, that helps digest food, along with its large counterpart.

SMALL INTESTINE

375. Fish whose skeletons are made primarily of this substance are known as chondrichthyes [kon-DRIK-tuh-eeze]. Those fish include rays and sharks. What is this type of tissue, which makes up much of the human nose and ears?

CARTILAGE

376. The strength of these storms is measured using a scale named for Pearson and Fujita. Those over lakes can be referred to as water spouts. Give the name for these storms, with funnel clouds, which occur frequently in their namesake alley.

TORNADO(ES)

377. This part of the electromagnetic spectrum has wavelengths just longer than X-rays. Too much of this type of radiation can lead to sunburn and skin cancer. Give the name for this type of electromagnetic radiation, with frequency just above violet light.

ULTRAVIOLET (RADIATION or WAVES or LIGHT) (Accept UV)

378. These photosynthetic organisms were once considered part of the protists. Types of these include red and brown, but technically not blue-green. Name these marine organisms, often a major source of food for fish.

ALGAE

379. This quantity is measured in the unit newtons. Types of this quantity include centripetal, normal, and magnetic. Name this phenomenon, any push or pull that affects an object's motion.

FORCE

380. Examples of these rocks include limestone and coal. Under heat and pressure, they can become metamorphic rock. Identify this type of rock, formed when minerals and organic materials clump together.

SEDIMENTARY (ROCK)

381. When this substance solidifies, it produces igneous rocks. Unlike magma, it is not contained underneath the Earth's surface. Name this substance, spewed from volcanoes.

LAVA

382. These scientists might study Sue at the Field museum. Unlike zoologists, they study fossils, rather than living organisms. Who are these scientists that study prehistoric life, usually by way of fossils?

PALEONTOLOGIST(S)

383. The Earth has magnetic and geographic versions of these. Usually they come in pairs, referred to as "north" and "south." What are these oppositely charged objects, such as the ends of a magnet?

POLE(S)

384. This quantity is the main topic of one of Newton's laws. Usually, it increases as mass and weight increase. What is this tendency of an object to resist changes in motion?

INERTIA

385. Examples of these pathogens include Ebola, HIV, and the common cold. Vaccines work to protect from infection with these. Name these pathogens that are smaller and simpler than bacteria.

VIRUS(ES)

386. This device debuted in April of 2010, and has had more than 250 million units sold. Varieties of it include the "air," the "mini," and the "pro." Identify this tablet, produced by Apple.

IPAD

387. These animals live almost entirely in the Southern Hemisphere. Varieties of these birds include the "little," "king," and "emperor." What are these flightless birds, which are excellent swimmers and inhabit Antarctica?

PENGUIN(S)

388. They are made up of a double layer of fatty molecules. Vesicles in a cell form from this structure. Name this structure that surrounds a cell.

(CELL) MEMBRANE

389. This kind of rock forms from air bubbles in volcanic rock. Water seeps inside the bubble, depositing minerals that collect inside. What kind of rock is hollow and filled with crystals?

GEODE

390. These non-organisms can only reproduce within the host cell that they infect. What agents consist of DNA in a protein shell?

VIRUS

391. Multiple answer required. one lays the eggs, a few do the fertilizing, and the rest do the work. What are the three kinds of bees in a typical bee hive?

QUEEN, DRONE, WORKER (any order)

392. These rounded instruments are usually marked with milliliter lines. What are these glass containers used to measure the volume of liquids?

GRADUATED CYLINDER(S)

393. You need a telescope to see these things that are labeled B and F, and a gap in these is known as the Cassini Division. What are these large circles of ice and rock surrounding our solar system's second largest planet?

(SATURN'S) RINGS (Accept RINGS OF SATURN)

394. They are known for having fronds and spores. Some of them have fiddleheads. What are these non flowering vascular plants?

FERNS

395. They are made in the nucleolus and line the rough E-R. What are these organelles, where proteins are created?

RIBOSOMES [RY-buh-zohms]

396. These organs work by extracting oxygen from water. What are these organs that fish use to breathe?

GILL(S)

397. An element's atomic number is based on the number of these particles. What are these positively charged particles, found in an atom's nucleus?

PROTON(S)

398. Types of these scientists include biomedical, chemical, and civil. What are these scientists called, who design structures and machines to solve problems in society?

ENGINEER(S)

399. The astronomical event known as "The Perseids" refers a shower of these objects. What are these small objects from outer space that enter Earth's atmosphere?

METEOR

400. The sweat glands and even parts of the lungs are parts of this system. What body system is responsible for the elimination of wastes from the body?

EXCRETORY (SYSTEM)

401. These chemicals help you feel good. What chemicals are produced by the central nervous system and pituitary gland when you exercise?

ENDORPHIN(S)

402. Members of this class of organisms, with a name meaning "head-foot," have excellent vision and advanced brains. What class includes the nautilus, squid and octopus?

CEPHALOPOD(IA) (Accept: CEPHALOPODS)

403. Two word answer. These contraptions change the magnitude or direction of a force. What contraptions include wedges, levers, and inclined planes?

SIMPLE MACHINE(S)

404. These objects convert chemical energy into electrical energy. What device, sometimes containing lithium, comes in D, triple A, and double A varieties?

BATTERIES

405. This flowing phenomenon is represented by the letter I in an equation and is measured in amperes. What do we call the flow of electrical charge, or the flow of electrons?

CURRENT

406. This phenomenon may occur when warm air lies above below-freezing air near the Earth's surface. What do we call transparent ice pellets, smaller than hail, that form as a result of the freezing of raindrops?

SLEET

407. One crystalline form of this element is known as graphite. What element helps make up all organic material, has atomic number 6 and symbol C, and composes coal and diamonds?

CARBON

408. A car will roll to a stop on its own because of this force. What force holds back the movement of a sliding or rolling object?

FRICTION

409. In aerodynamics, this force is generated by the wings of an airplane. What force keeps an airplane suspended in the air?

LIFT

410. Mammals are named for these glands. What glands produce milk in mammals?

MAMMARY (GLANDS)

411. Edward Jenner's work has led to the announcement by the World Health Organization that this disease is eradicated. What infectious disease was known as the red plague?

SMALLPOX

412. Frogs lay their eggs in water in this clump. What is the name of a floating clump of frog eggs?

(FROG) SPAWN

413. When a wave moves through the water, this is the point of maximum downward displacement. What is the name of the lowest point on a wave, the opposite of the crest?

TROUGH

414. During this stage of metamorphosis, wings begin to form. What is the stage of development where butterflies will not eat or move?

PUPA

415. One example of this kind of meteorological event was a super one called Haiyan that devastated the Philippines in 2013. What is the term for a tropical cyclone with wind speeds over 74 miles per hour that occurs over the Northwest Pacific Ocean?

TYPHOON (Not: HURRICANE)

416. The phase change that occurs at this temperature for a substance is usually the result of an increase in pressure or heat. What is the term for the temperature at which a solid changes to a liquid?

MELTING POINT

417. More efficient than filament bulbs, this light source gets its name from its twin terminal structure. What is the three-word name of the light source, which generates less heat than incandescent bulbs, and is abbreviated LED?

LIGHT EMITTING DIODE

418. In an insect, it lies below the thorax. What is this body region, which for humans includes the stomach and other digestive organs?

ADBOMEN

419. In animals, this system is responsible for the release of carbon dioxide and intake of oxygen. What is this body system that, in humans, is comprised of the lungs and other organs?

RESPIRATORY (SYSTEM)

420. This muscle surrounds and controls light intake to the pupil. What is this colored part of the human eye?

IRIS

421. This chemical compound is combined with water, in the presence of sunlight, in order to produce sugar and oxygen in photosynthesis. What is this compound, with symbol C-O-two?

CARBON DIOXIDE

422. This medical instrument consists of flexible rubber tubes that attach a chest piece to metal tubes ending in earpieces. What is this device used to hear internal sounds like stomach gurgles, breaths, and heartbeats?

STETHOSCOPE

423. This element is the most common element in Earth's atmosphere. What is this element, with atomic number seven and symbol N?

NITROGEN

424. Plants absorb this through their chloroplasts [KLOR-oh-plasts], where its energy is used to react water with carbon dioxide. What is this energy source that is necessary for photosynthesis?

(SUN) LIGHT

425. In an experiment, the scientist manipulates the "independent" type of this. What is this feature of an experiment or scientific study, examples of which could be mass, height, or speed?

VARIABLE

426. Examples of this state of matter include the Northern Lights and much of what makes up stars. What is this fourth state of matter, which is hotter than normal gas?

PLASMA(S)

427. These organisms are invertebrates with no limbs, and they have simple organ systems. What is this group of organisms, which includes the "tape," "round," and "earth" kinds?

WORM(S)

428. This atmospheric layer has been getting thinner, primarily due to man-made chemicals. What is this layer of the atmosphere, made of three bonded oxygen atoms?

OZONE (LAYER)

429. Steps in it include defining a research question, the hypothesis, and experimentation. What is this method for conducting research in fields like biology and chemistry?

SCIENTIFIC (METHOD)

430. This molecule is broken down during the first step of cellular respiration, and is carried in human blood. What is this molecule, a simple form of sugar?

GLUCOSE

431. This organ connects your brain to your body. What is this organ, which differentiates vertebrates from invertebrates?

SPINAL CORD

432. Examples of this weather event include sleet, snow, and rain. What is this part of the water cycle, in which water returns to the earth from clouds?

PRECIPITATION

433. This process is separated into the phases G1, S, and G2. What is this phase of cell division, which occurs in-between periods of active cell division?

INTERPHASE

434. Two word answer please. It happens when the moon passes directly behind the Earth, into its shadow. What is this phenomenon that makes the moon disappear from sight?

LUNAR ECLIPSE

435. This tissue is made of mostly undifferentiated cells, and gives rise to tissues such as the phloem and xylem. What is this plant tissue that comes in unifacial and vascular types?

CAMBIUM

436. This term describes the duplication and distribution of chromosomes. What is this process of cell duplication in which one cell gives rise to two identical daughter cells?

MITOSIS [mahy-TOH-sis]

437. This process typically takes place alongside erosion. What is this process, by which wind, water, and ice break down exposed objects such as rock?

WEATHERING

438. This is an essential characteristic of vertebrates. What is this rod made of cartilage that is found in all vertebrate embryos?

NOTOCHORD

439. It typically involves a rope, threaded around a wheel. What is this simple machine, often used to lift objects?

PULLEY(S)

440. Two word answer please. This arachnid [uh-RACK-nid] has a dark mark in the shape of a violin and often avoids humans . What is this spider found in the southern and western United States that is venomous and usually tan in color?

BROWN RECLUSE

441. This subatomic particle is nearly without mass compared with other subatomic particles. What is this stable particle, which carries a negative charge?

ELECTRON

442. Flying along with these currents can allow airplanes to reduce their travel time. What is this term for these fast flowing air currents in Earth's atmosphere?

JET STREAM

443. When this matter comes in contact with normal matter, annihilation occurs. What is this type of matter, not to be confused with dark matter?

ANTIMATTER

444. In its native habitat, the roots or other parts of this plant survive year-round, even after reproduction. What is this type of plant that, unlike annuals, can survive for multiple years without being replanted?

PERENNIAL (PLANT)

445. This tissue takes water and nutrients up from the roots. What is this type of vascular tissue, which complements the phloem?

XYLEM [ZYE-lehm]

446. One thousand of these make the basic SI unit of mass. What is this unit of measure, once based on the density of water?

GRAM(S)

447. This type of plant tissue takes nutrients from leaves to the rest of the plant. What is this vascular tissue, which works in conjunction with the xylem?

PHLOEM [FLO-um]

448. This is formed from a thick layer of condensed water droplets in the lower atmosphere or surface of earth. What is this weather phenomenon, which mixes with smoke to form smog?

FOG

449. An example of this kind of form for carbon is used to determine the age of bones. What kind of form has fewer or more neutrons than the average atom?

ISOTOPE

450. This phenomenon is the mechanism by which objects become permanently magnetized. What kind of magnetism is generated spontaneously by materials like cobalt, nickel, and iron?

FERROMAGNETISM (Accept word forms)

451. This quantity can be measured in feet per second or miles per hour. What measure tells how fast an object is travelling, with distance divided by time?

SPEED (ACCEPT: VELOCITY)

452. Examining these ancient objects can give scientists clues about life millions of years ago. What objects found in sedimentary rock are remnants of dead creatures and are studied by paleontologists?

FOSSILS

453. This organ has a large surface area and must have water continuously flowing over it to extract oxygen from water. What organ allows aquatic animals, such as fish, to breathe?

GILL

454. This organ sits behind the stomach and produces both enzymes to aid digestion and many hormones. What organ regulates the levels of insulin in the body?

PANCREAS

455. These organisms can often be found growing in dark, moist environments, and come in types like club and sac. What organisms include molds and mushrooms?

FUNGI

456. These characteristic body parts can be used to differentiate moths from butterflies. What parts of the butterfly have clubs on the end, while those of moths do not?

ANTENNAE

457. Two word answer required. What phase of the moon occurs when over one-half is visible, and it is increasing toward a full moon?

WAXING GIBBOUS [GIB-us]

458. Objects with this physical characteristic do not cast a shadow. What physical characteristic is defined as an object's ability to allow light to completely pass through it?

TRANSPARENCY (Accept word forms)

459. This part of a plant bears the buds and shoots, and usually ends at the roots. What plant part supports the plant and transports water and nutrients from the roots to the leaves?

STEM

460. This quantity involves displacement, and is measured in joules. What quantity is the product of force times distance?

WORK

461. These rocks are made from bubbles in volcanic rock or burrows with silica-containing water on the inside. What rocks are hollow with crystals lining the inside?

GEODES

462. This concept is also called the Archimedes Principle. What scientific concept allows low density objects to float in water?

BUOYANCY

463. His father's work in construction played an important role in this scientist's work. What scientist, for whom a special prize is named, invented dynamite in 1867?

(ALFRED) NOBEL

464. These cells found in sponges capture spermatozoa [spur-mat-uh-ZOH-uh] released by other sponges. What sponge cells have a nucleus, a collar, and a flagellum?

CHOANOCYTES [koh-AN-uh-sights] (Accept: COLLAR CELLS)

465. Famous ones include Edna, Ike, and Katrina. What storms see winds exceeding 74 miles per hour?

HURRICANES

466. If birds didn't have this specific type of skeletal feature, they wouldn't be able to fly. What structural adaptation results in lighter bodies, and allows birds to fly long distances?

HOLLOW BONES (accept like answer)

467. Many plant cells contain this pigment that allows photosynthesis to occur. What substance causes the green color in plants and can be found in the chloroplast of the cell?

CHLOROPHYLL

468. Crops grow best on this type of land. What term beginning with the letter A describes land that is capable of being farmed?

ARABLE

469. Unlike plants, this type of organism cannot make its own food. What term, beginning with the letter H, is used to describe organisms that feed off other organisms for energy?

HETEROTROPHIC

470. This event occurs when the Earth's upper atmosphere scatters and reflects sunlight. What time occurs in the morning or the evening when the sky is barely lit with the sun's rays?

TWILIGHT

471. The most common type of this device looks like a triangular cube. What transparent device separates white light into a spectrum of colors?

PRISM

472. No matter if it's a hot summer day, or cold winter night, the body temperature of mammals and birds stays about the same. What type of animal maintains a constant core body temperature, despite its surrounding environment?

WARM BLOODED (ACCEPT: ENDOTHERM OR ENDOTHERMIC)

473. In order to reduce the dangers of disease, communities may spray insecticides to kill these insects. What type of pesky insect can cause diseases like malaria?

MOSQUITO

474. Each year advertisements are created to advise people about the importance of this inoculation. What type of vaccine is usually given each fall to prevent its namesake disease?

FLU (SHOT or VACCINE) (Accept: INFLUENZA)

475. This unit is measured in joules [jewels] per second. What unit of power is used to measure electrical devices like lightbulbs?

WATT

476. Marie Curie invented this word when she studied uranium and thorium. What word, based on the Latin word for "ray", refers to the particles emitted from the nuclei of atoms?

RADIOACTIVITY (Accept word forms)

477. Gravitational potential energy is equal to this quantity times gravity, times mass. When compared to sea level, this quantity is called altitude. Give this term used for the distance from the bottom to the top of an object.

HEIGHT

478. In one of Newton's laws, force is equal to this quantity, times acceleration. When multiplied by gravity, it is the same as weight. Give this term for the amount of a substance present.

MASS

479. This is one of the main types of soil. When this type of soil is moistened, it will feel slick, and it will roll up easily and form a ball. What is this type of soil, that can turn hard and compact during the summer months?

CLAY

480. These body organs contain thousands of tiny alveoli. When you get an infection in these organs, it can lead to pneumonia. Name these primary organs of the respiratory system.

LUNG(S)

481. Alpha Centauri is the closest one of these astronomical bodies to our solar system. Which astronomical bodies are made up of gases, have nuclear reactions, and can be seen from Earth as points of light?

STARS

482. Computers do not cause viruses in humans, but these body parts can be hurt by spending too much time on the computer. Which body parts can suffer from carpal tunnel disorder?

WRIST(S)

483. Pioneers in this branch of science include Einstein, Marie Curie, and Newton. Which branch of science studies the interaction between matter and energy, and the way forces act on the environment?

PHYSICS

484. Thousands of cells from this layer rub off your body every day. Which layer of skin is the thinnest, and contains dead cells on its outermost section?

EPIDERMIS

485. Galileo discovered this dry, volcanic moon in 1610. Which of Jupiter's Galilean moons has the closest orbit to Jupiter?

IO [EYE-oh]

486. This law explains how the thrust of a rocket works, or why we sometimes get whiplash after being rear-ended. Which of Newton's Laws suggests that every action has an equal and opposite reaction?

(NEWTON'S) THIRD (LAW OF MOTION) (Accept: 3)

487. Objects in this region include Vesta, Pallas, and Ceres. Which region of our solar system contains masses of rock and lies between Mars and Jupiter?

ASTEROID BELT

488. These subatomic particles have almost no mass, and do not contribute to an atom's atomic number. Which subatomic particles exist in a cloud surrounding the core of the atom and have a negative charge?

ELECTRONS

489. The star referred to as Proxima is normally considered part of this star system. While once thought to be a single star, it is actually a binary pair. What is this closest star system to our own solar system?

ALPHA CENTAURI

490. Topics these scientists might study include stratification, continental drift, and the Mohs Scale. Who are these scientists that study rocks, minerals, and other matter that make up the earth?

GEOLOGIST(S)

491. She said, "It is not easy to be a pioneer - but oh, it is fascinating!" Who became the first female to earn a medical degree in the United States?

(ELIZABETH) BLACKWELL

492. This botanist published multiple bulletins on the uses of sweet potatoes. Who is this American scientist and inventor, known for discovering countless uses for the peanut?

(GEORGE WASHINGTON) CARVER

493. This is one of the latest kinds of printers. Wolowitz in Big Bang Theory used one of these to print a copy of himself. What is this printer that can print copies of toys, for instance?

3-D (PRINTER)

494. This term is used to describe a moon that is more than half, but less than fully illuminated. You might say it is the opposite of a crescent moon. What is this term that may be preceded by the words "waxing" or "waning"?

GIBBOUS [GIB-uhs]

495. This phenomenon is contrasted with reflection and diffraction. You see it when you place an object in water--the object appears broken or bent. What is this term for the bending or changing direction of light?

REFRACTION

496. What substance, created by the evaporation of water, causes humidity in the air?

(WATER) VAPOR

497. What nutritional group is the main source of energy for your body and includes sugar, starch or fiber?

CARBOHYDRATES (Accept: CARBS)

498. What type of sedimentary rock contains tiny pebbles cemented together by sand or mud?

CONGLOMERATE (ROCK)

499. What term beginning with "F" is used to describe a developing mammal, or the unborn offspring of vertebrates after the embryonic stage?

FETUS

500. What form of cancer begins in the blood and bone marrow, and results in high numbers of abnormal white blood cells?

LEUKEMIA